Audit Trail: Case 11: Shoulder closure on divided highway

Field	Value
Plan status	1 change · 3 needs attention · 15 checked · 2 pending · reference
MUTCD typical application	TA-3
CDOT standard sheet	S-630-1
Case	Case 11: Shoulder closure on divided highway
Taper length	50 ft (L/3 (shoulder taper))
Buffer space	200 ft
Device spacing: taper	30 ft
Device spacing: tangent	60 ft
Crew steps	14
Routing	shoulder_no_reduction

Guessed inputs

- Street class: Arterial. Guessed from the road (OSM highway=primary). The operator did not confirm it. The jurisdiction's adopted classification map governs.
- Jurisdiction: Denver. Guessed from the pin (Pin is inside Denver municipal limits (US Census TIGER/Line Place boundaries, 2025 vintage).). The operator did not confirm it. Confirm the jurisdiction with the permitting authority.
- Road type: Urban street. Guessed from the road and nearby places (OSM highway=primary; urban, inferred from nearby places). The operator did not confirm it. Confirm on site whether the work is in an urban or rural setting.

Taper Length

Speed 30 mph < 45 mph threshold -> using L = W x S^2 / 60

L = 10 x 30^2 / 60 = 150 ft

L/3 = 150 / 3 = 50 ft

Field	Value
Closure type	shoulder
Offset (shoulder width)	10 ft (assumed)
Required taper (L/3 (shoulder taper))	50 ft

Source: MUTCD 11th Ed. Sec 6B.08, Table 6B-4 (taper length L). Shoulder closures use L/3 per Sec 6B.08 (Table 6B-3).

CDOT reference: CDOT S-630-1 Case 11 (right-shoulder closure on divided highway)

Buffer Space

MUTCD Table 6B-2: 200 ft (CDOT supplement silent at this speed)

Source: MUTCD 11th Ed. Sec 6B.06, Table 6B-2 (stopping sight distance)

Channelizing Device Spacing

30 mph -> 30 ft spacing (MUTCD Sec 6K.01: spacing equals speed in feet)

30 mph -> 60 ft spacing (MUTCD Sec 6K.01: spacing equals 2x speed in feet)

L/3 = 50 ft / 30 ft max spacing -> 3 drums at 25.0 ft intervals

1000 ft / 60 ft max spacing -> 18 cones at 58.8 ft intervals

Buffer: 200 ft / 60 ft max spacing -> 3 cones at 50.0 ft intervals (CDOT S-630-1 Sheet 2 General Note 18b: tangents along the buffer space)

Downstream taper: 50 ft at ~25 ft spacing -> 2 cones (MUTCD Sec 6B.08: 50-100 ft downstream taper); 18 tangent + 3 buffer + 2 downstream = 23 cones total

Source: MUTCD 11th Ed. Sec 6K.01

Advance Warning Signs

Speed 30 mph, road_type = 'urban_low' -> Table 6B-1 category: urban_low

A = 100 ft, B = 100 ft, C = 100 ft

Position	Code	Station (ft)	Distance from Taper (ft)
C (furthest)	W20-1	1,550	300 upstream
B (middle)	W20-2	1,450	200 upstream
A (nearest)	W21-5aR	1,350	100 upstream

Source: MUTCD 11th Ed. Table 6B-1

Colorado Requirements (CDOT S-630-1)

Check	Result	Citation	Detail
Signs on both sides of divided highway	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 8	Not required: one shoulder closed (Note 8's Case 11 exception). Signs placed: 6 left, 8 right.
G20-5P Work Zone signs every 2,640 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 4	Zone length: 1,550 ft. Required: 1. Placed: 1.
Speed reduction <= 15 mph per sign installation	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 3	No work-zone speed reduction. Posted speed 30 mph applies throughout the zone.
Flagger station lighting 500W @ 8 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 22	Not applicable (no flaggers).

AADT threshold for mobile operations (<= 2,000): Not applicable (not a mobile operation). (CDOT S-630-1 (July 2026) Sheet 23, Case 38 Note 1)

All Colorado checks pass: True

CDOT Case Reference

Case 11: Shoulder closure on divided highway

This scenario matches CDOT Standard Plan S-630-1, Case 11: shoulder closure on a divided highway.

The generated plan follows the same device layout as the reference case with spacing computed for 30 mph.

Routing: shoulder_no_reduction

Reference:
https://www.codot.gov/safety/traffic-safety/assets/s-standard-plans/s-630-1/S-630-01%20(19-Page%20Set).pdf

Geometry Validation

Work zone 1000 ft vs required taper 50 ft (L/3 (shoulder taper)) + buffer 200 ft.

All geometry checks pass: True

Source: MUTCD 11th Ed. Sec 6B.06 (buffer) and Sec 6B.08 (taper)

Corridor Validation

OpenStreetMap corridor check ran with no warnings.

Site Conditions

Scanned along the corridor against OpenStreetMap at 2026-10-18T02:32:31+00:00 (3420 ms, memoised, via https://overpass.openstreetmap.fr/api/interpreter). The five rule-bearing conditions below are plan facts; a detected condition fired the matching Site Adjustment that follows.

Condition	Result	Evidence
Adjacent at-grade intersection	DETECTED	42 found · nearest 64 ft from anchor · unnamed at 39.7400, -104.9648 [work_zone @ 666 ft]
Adjacent interchange (highway ramps)	None along the corridor	
Pedestrian sidewalks	DETECTED	46 found · nearest 65.9 ft from anchor · unnamed at 39.7401, -104.9678 [lateral 63 ft off centerline]
Bike lane / cycleway	None along the corridor	
School zone	None along the corridor	
Railroad crossing	Reference: none	
Hospital	Reference: none	
Road curvature	Reference: none	

Site Adjustments

Site-condition flags layered onto the baseline MUTCD/CDOT layout. Each adjustment is traced to its source rule; the rendered plan, device list, and crew narrative reflect every item below.

Rule	Adjustment
MUTCD §6N.12 p. 848: Work within the Traveled Way at an Intersection (11th Ed.)	No devices added: the cross-street approach layout isn't generated. See the pending-verification disclosure.
MUTCD §6C.02: pedestrian considerations in work zones	Added 2 Type III barricades (sidewalk closure points) and 2 R9-9 SIDEWALK CLOSED signs at the upstream and downstream ends of the work zone.

Flagger Control

Not applicable for this scenario.

Source: MUTCD 11th Ed. Chapter 6D (Flagger Control) and Chapter 6E (One-Lane, Two-Way Traffic Control).

Pending Verification

1 reference(s) pending verification. An adjacent intersection is flagged, but Conestruct does not generate the cross-street approach layout or evaluate traffic-signal operation, and adds no devices for it. Cross-street control design and the signal-operation review (MUTCD §6N.12, 11th Ed. p. 848; Part 4 for signal-head visibility) remain with the traffic control supervisor. Intersection support is tracked at issue #117; corner-quadrant work at issue #128. Tracking: https://github.com/rtmakatura/conestruct/issues/117

- [intersection_layout_not_generated] An adjacent intersection is flagged, but Conestruct does not generate the cross-street approach layout or evaluate traffic-signal operation, and adds no devices for it. Cross-street control design and the signal-operation review (MUTCD §6N.12, 11th Ed. p. 848; Part 4 for signal-head visibility) remain with the traffic control supervisor. Intersection support is tracked at issue #117; corner-quadrant work at issue #128. (https://github.com/rtmakatura/conestruct/issues/117)